

## Demonstration of Proposed Features

|               |                                                                |
|---------------|----------------------------------------------------------------|
| Date          | 15 July 2026                                                   |
| Project Name  | Rising Waters: A Machine Learning Approach to Flood Prediction |
| Maximum Marks | 1 Mark                                                         |

This document captures all the features proposed during planning for Rising Waters and tracks whether each was successfully implemented and demonstrated.

| S<br>·<br>N<br>o | Feature Name                    | Description                                                               | Status      | Demonstrated | Remarks                                                                                                 |
|------------------|---------------------------------|---------------------------------------------------------------------------|-------------|--------------|---------------------------------------------------------------------------------------------------------|
| 1                | Flood Risk Prediction (XGBoost) | Predicts flood / no-flood risk from 5 rainfall and cloud-cover features   | Implemented | Yes          | Core deliverable; tested with multiple safe and flood-risk scenarios                                    |
| 2                | Rainfall Input Form             | Collects Cloud Cover, Annual Rainfall, and three seasonal rainfall values | Implemented | Yes          | Validated on both client and server side                                                                |
| 3                | Two-Layer Input Validation      | Client-side JS checks plus server-side Python range validation            | Implemented | Yes          | Confirmed via boundary and invalid-input test cases                                                     |
| 4                | Animated Result Visualization   | Circular probability gauge with a clear risk / safe verdict page          | Implemented | Yes          | Matches the final deployed UI                                                                           |
| 5                | Model Training Pipeline         | Auto-detects real vs. synthetic data, trains and saves model artifacts    | Implemented | Yes          | train.py runs end-to-end without manual intervention                                                    |
| 6                | Live Public Deployment          | Hosted application accessible via a public URL                            | Implemented | Yes          | Live at <a href="https://rising-waters-12fj.onrender.com/">https://rising-waters-12fj.onrender.com/</a> |

### Feature Implementation Summary

|                                 |      |
|---------------------------------|------|
| Total Features Proposed         | 6    |
| Total Features Implemented      | 6    |
| Total Features Demonstrated     | 6    |
| Overall Implementation Rate (%) | 100% |